
WASSERSTEIN GRADIENT FLOWS FOR MEAN-FIELD MARKET MAKING

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ABSTRACT

Financial markets do not literally minimize a physical energy, but the distribution of strategic agents in a market may evolve toward equilibria that minimize a cost-like *free-energy* functional. This project studies a mean-field market-making model in which the population distribution of dealer inventories evolves under the combined effect of an inventory/risk penalty, a crowding interaction, and an entropic dispersion term. We formulate the Fokker–Planck side of the population dynamics as a Wasserstein gradient flow whenever the interaction structure is *potential*, and we discretize the evolution with a Jordan–Kinderlehrer–Otto (JKO) proximal scheme. The aim is to test whether this variational formulation yields a stable and interpretable numerical method for computing market-making equilibria, and to quantify how risk aversion, crowding, and exogenous liquidity shocks reshape the distribution of inventories. We are careful to delimit the scope of the analogy: a general mean-field game need *not* have gradient-flow structure, and we identify the additional hypotheses under which the JKO machinery is rigorously available.

1 Introduction

A recurring intuition imported from physics into quantitative finance is that “systems evolve toward states of lower energy.” Taken literally at the level of prices or returns, the analogy fails: raw asset returns are heavy-tailed, nonstationary, and carry no natural notion of a scalar energy that they decrease over time. The object that *does* admit such a description is not the price but the *distribution of agent states* — dealer inventories, beliefs, risk exposures, or order-flow intensities — which can plausibly relax toward an equilibrium that minimizes a cost-like functional. The natural geometry for this relaxation is not Euclidean but the Wasserstein geometry of optimal transport, in which a distribution evolves by transporting probability mass at minimal kinetic cost.

This shift of viewpoint is what makes the physical analogy precise. The seminal observation of ? is that the Fokker–Planck equation can be written as the gradient flow, *in the 2-Wasserstein metric*, of a free-energy functional: a density ρ_t moves in the direction of steepest descent of an energy $\mathcal{F}$, subject to the cost of transporting mass. Their variational time-discretization — now called the JKO scheme — replaces the PDE by a sequence of minimizations of “energy plus transport cost,”

$$\rho_{k+1} \in \arg \min_{\rho \in \mathcal{P}_2(\mathbb{R}^d)} \left\{ \mathcal{F}(\rho) + \frac{1}{2\tau} W_2^2(\rho, \rho_k) \right\}, \quad (1)$$

which is exactly the statement “the next state is a low-energy state, penalized for moving too far too fast.” This is the structure we wish to exploit on the finance side.

Why market making. Mean-field games (MFGs) provide a setting in which a distribution of strategic agents genuinely evolves toward equilibrium. In the market-making model of ?, a market maker quotes against a continuum of strategic market takers whose aggregate behavior enters through a mean-field interaction; the equilibrium is characterized by a coupled Hamilton–Jacobi–Bellman (HJB) and Fokker–Planck system. The Fokker–Planck half of that system governs precisely the population density that our free-energy picture is meant to describe. Inventory is the natural state variable:

market makers dislike large positions, similar quoting behavior across dealers produces crowding, and exogenous order flow tilts the landscape that the inventory distribution relaxes within. Each of these effects has a clean counterpart in a free-energy functional, which is what makes the analogy concrete rather than decorative.

The central caveat. It is tempting to assert that “the Fokker–Planck half of an MFG is a Wasserstein gradient flow.” This is too strong. In a general MFG the forward equation is a continuity equation driven by the optimal feedback control coming from the HJB equation,

$$\partial_t m - \nabla \cdot (m D_p H(x, \nabla u, m)) - \nu \Delta m = 0, \quad (2)$$

and such a drift need not be the gradient of the first variation of any scalar energy. The Fokker–Planck equation becomes a bona fide Wasserstein gradient flow only under additional structure — for instance potential (variational) MFG structure, entropic regularization, or convex costs whose induced drift can be written as $-\nabla \delta \mathcal{F} / \delta m$. We therefore do not claim that mean-field market making is “automatically JKO.” We instead pose the design question:

Remark 1 (Scope of the analogy) *We seek to identify or design a market-making mean-field game whose population dynamics admit a Wasserstein-gradient-flow (equivalently, a generalized JKO) formulation, and to use that structure for stable computation and for interpretability of the resulting equilibria. The energy lives on the distribution of agent states, never on raw prices or returns.*

Contributions. Within this scope, the project proceeds as follows.

1. We formulate a free-energy functional $\mathcal{F}$ on the distribution of dealer inventories that combines an inventory/risk potential, a pairwise crowding interaction, and an entropic term, and we record the precise conditions under which the associated Fokker–Planck dynamics are the W_2 -gradient flow of $\mathcal{F}$ (Section 2).
2. We discretize the flow with the JKO scheme (1) and benchmark it against a standard Fokker–Planck PDE solver, comparing energy dissipation, conservation of mass and positivity, and convergence to the predicted equilibrium.
3. We study the comparative statics of the equilibrium inventory distribution as a function of risk aversion and crowding, and its transient response to exogenous liquidity shocks modelled as time-dependent tilts of the potential.

Related work. The variational formulation of the Fokker–Planck equation is due to ?; the metric-geometric theory of gradient flows in the space of probability measures is developed in ?, with the formal Riemannian (“Otto”) calculus introduced in ? and surveyed in ?. The mean-field game framework originates with ?; see ? for the coupled HJB–Fokker–Planck system and the notion of potential games. Our market-making model follows the strategic mean-field formulation of ?, itself built on the inventory-control approach of ?. On the algorithmic side, ? analyze JKO-type proximal descent in Wasserstein space and establish Kullback–Leibler guarantees for JKO-flow models under suitable assumptions, which informs our choice of numerical scheme. Finally, Wasserstein distances have been used directly for regime and change-point detection in high-dimensional series [?]; we return to this as an honest, structure-light alternative to monitoring a JKO residual.

Outline. Section 2 collects the required preliminaries: optimal transport and the Wasserstein space, free-energy functionals and their Wasserstein gradient flows, the JKO minimizing-movement scheme, the mean-field-game system and the potential structure that links the two, and the market-making model that fixes our notation.

2 Preliminaries

Throughout, $\mathcal{P}_2(\mathbb{R}^d)$ denotes the set of Borel probability measures on $\mathbb{R}^d$ with finite second moment, $\int_{\mathbb{R}^d} |x|^2 d\mu(x) < \infty$. Densities are written ρ, m and identified with the corresponding measures when absolutely continuous with respect to Lebesgue measure. We write $f * \rho$ for convolution and $D_p H$ for the gradient of a Hamiltonian H in its momentum slot.

2.1 Optimal transport and the Wasserstein space

Definition 2 (2-Wasserstein distance) *For $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$, the squared 2-Wasserstein distance is*

$$W_2^2(\mu, \nu) = \inf_{\gamma \in \Pi(\mu, \nu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} |x - y|^2 d\gamma(x, y), \quad (3)$$

where $\Pi(\mu, \nu)$ is the set of couplings of μ and ν , i.e. measures on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ and ν .

The metric space $(\mathcal{P}_2(\mathbb{R}^d), W_2)$ is the central object of the theory [??]. Its dynamic, fluid-mechanical characterization is due to ?:

$$W_2^2(\mu, \nu) = \inf_{(\rho_t, v_t)} \int_0^1 \int_{\mathbb{R}^d} |v_t(x)|^2 \rho_t(x) dx dt, \quad \partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0, \quad \rho_0 = \mu, \quad \rho_1 = \nu, \quad (4)$$

the infimum being over curves $(\rho_t)_{t \in [0,1]}$ in $\mathcal{P}_2$ joined by a velocity field v_t through the continuity equation. Equation (4) endows $\mathcal{P}_2$ with a formal Riemannian structure [?]: tangent vectors at ρ are represented by velocity potentials ϕ via $\partial_t \rho = -\nabla \cdot (\rho \nabla \phi)$, with metric $\langle \partial_t^{(1)} \rho, \partial_t^{(2)} \rho \rangle_\rho = \int \nabla \phi_1 \cdot \nabla \phi_2 \rho dx$. This is the geometry in which the gradient flows below are defined.

2.2 Free-energy functionals and Wasserstein gradient flows

We consider free-energy functionals of ? type, comprising a potential, an interaction, and an internal (entropic) term:

$$\mathcal{F}(\rho) = \underbrace{\int_{\mathbb{R}^d} V(x) \rho(x) dx}_{\text{potential energy}} + \underbrace{\frac{1}{2} \iint_{\mathbb{R}^d \times \mathbb{R}^d} W(x, y) \rho(x) \rho(y) dx dy}_{\text{interaction energy}} + \underbrace{\beta \int_{\mathbb{R}^d} \rho(x) \log \rho(x) dx}_{\text{internal energy / entropy}}, \quad (5)$$

with V a confining potential, W a symmetric interaction kernel, and $\beta > 0$. The *first variation* of $\mathcal{F}$ at ρ , denoted $\frac{\delta \mathcal{F}}{\delta \rho}$, is the function whose perturbation gives the directional derivative; for (5),

$$\frac{\delta \mathcal{F}}{\delta \rho}(x) = V(x) + (W * \rho)(x) + \beta (\log \rho(x) + 1). \quad (6)$$

Definition 3 (Wasserstein gradient flow) A curve $(\rho_t) \subset \mathcal{P}_2(\mathbb{R}^d)$ is a gradient flow of $\mathcal{F}$ in the W_2 geometry if it solves

$$\partial_t \rho_t = \nabla \cdot \left(\rho_t \nabla \frac{\delta \mathcal{F}}{\delta \rho}(\rho_t) \right). \quad (7)$$

Substituting (6) into (7) yields the nonlinear Fokker–Planck (McKean–Vlasov) equation

$$\partial_t \rho_t = \nabla \cdot (\rho_t \nabla V) + \nabla \cdot (\rho_t \nabla (W * \rho_t)) + \beta \Delta \rho_t, \quad (8)$$

which makes the physical reading explicit: the distribution drifts downhill in the potential V , is repelled from regions of high crowding through $W * \rho$, and is dispersed by the entropic Laplacian. Along (7) the energy is dissipated at rate $\frac{d}{dt} \mathcal{F}(\rho_t) = - \int \rho_t |\nabla \frac{\delta \mathcal{F}}{\delta \rho}|^2 dx \leq 0$, the finance analogue of relaxation toward equilibrium.

2.3 The JKO minimizing-movement scheme

The variational time-discretization of ? approximates (7) by the proximal iteration

$$\rho_{k+1} \in \arg \min_{\rho \in \mathcal{P}_2(\mathbb{R}^d)} \left\{ \mathcal{F}(\rho) + \frac{1}{2\tau} W_2^2(\rho, \rho_k) \right\}, \quad \rho_0 = \bar{\rho}, \quad (9)$$

with step size $\tau > 0$. As $\tau \rightarrow 0$ the piecewise-constant interpolation of (ρ_k) converges to the gradient flow (7) [??]; the scheme is the Wasserstein analogue of the implicit (backward) Euler method, and inherits its unconditional stability, automatic preservation of nonnegativity and total mass, and monotone energy decrease $\mathcal{F}(\rho_{k+1}) \leq \mathcal{F}(\rho_k)$. Proximal descent of this form admits exponential-convergence and Kullback–Leibler guarantees under convexity-type assumptions on $\mathcal{F}$ [?], which is what recommends (9) as a numerical method rather than only a theoretical device.

2.4 Brenier’s theorem and the transport map between JKO iterates

The iteration (9) produces the sequence (ρ_k) as a list of distributions, but it says nothing directly about *where mass goes* from one step to the next. Brenier’s theorem supplies exactly this missing object: under a mild regularity assumption it upgrades the static coupling of (3) to a deterministic map, so that each JKO step is realized by transporting the current distribution through a single function of the state.

Theorem 4 (Brenier [??]) Let $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$ and suppose μ is absolutely continuous with respect to Lebesgue measure. Then the infimum in (3) is attained by a unique coupling, and this coupling is induced by a μ -almost-everywhere unique map $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$; that is, the optimal γ equals $(\text{id}, T)_\# \mu$ and

$$\nu = T_\# \mu, \quad W_2^2(\mu, \nu) = \int_{\mathbb{R}^d} |x - T(x)|^2 d\mu(x). \quad (10)$$

Moreover $T = \nabla\varphi$ for a convex potential $\varphi : \mathbb{R}^d \rightarrow \mathbb{R}$, and if ν is also absolutely continuous then T is invertible μ -almost everywhere with inverse $T^{-1} = \nabla\varphi^*$ (the map $\nabla\varphi^*$ being the Brenier map from ν to μ).

Here $T_{\#}\mu$ denotes the pushforward, $(T_{\#}\mu)(A) = \mu(T^{-1}(A))$ for measurable A : the distribution obtained by moving each sample $x \sim \mu$ to $T(x)$. The content of the theorem is that among all couplings, the W_2 -optimal one is not a genuinely stochastic assignment but a function, and that function is the gradient of a convex potential.

The step-to-step transport. We apply Theorem 4 with $\mu = \rho_k$ and $\nu = \rho_{k+1}$. By Theorem 9(i)–(ii) and Theorem 4 each JKO iterate is absolutely continuous with a strictly positive density, so both hypotheses hold at every step and there is a unique Brenier map T_k with

$$\rho_{k+1} = (T_k)_{\#}\rho_k, \quad T_k = \nabla\varphi_k, \quad \varphi_k \text{ convex}, \quad (11)$$

together with its inverse T_k^{-1} carrying ρ_{k+1} back to ρ_k . The map T_k is the deterministic realization of one proximal step: rather than recording only that the distribution changed from ρ_k to ρ_{k+1} , it records the displacement $x \mapsto T_k(x)$ that each unit of mass undergoes. Iterating, the composition $T_{k-1} \circ \dots \circ T_0$ transports the initial datum ρ_0 onto ρ_k , so the whole discrete flow is a composition of gradient-of-convex maps.

Consistency with the Otto geometry. The map form is the discrete-time image of the velocity picture of Section 2.1. In the continuous flow (7), mass at x moves with velocity $v_t(x) = -\nabla \frac{\delta\mathcal{F}}{\delta\rho}(\rho_t)(x)$; over a step of size τ the Brenier map is, to first order, the corresponding displacement,

$$T_k(x) = x - \tau \nabla \frac{\delta\mathcal{F}}{\delta\rho}(\rho_{k+1})(x) + o(\tau), \quad (12)$$

which is the fully-implicit (backward) reading of (9): the displacement $T_k - \text{id}$ is proportional to the Wasserstein gradient evaluated at the *new* iterate, scaled by τ . Equation (12) is what identifies $(T_k - \text{id})/\tau$ with the W_2 -gradient direction and underlies the alignment diagnostic of Section 5.

Why the map is the object to learn. Because the transport determines ρ_{k+1} from ρ_k through (11), knowing T_k is equivalent to knowing the distribution shift, and its inverse predicts the shift in the reverse direction. This is what makes the map, rather than the pair of densities, the natural quantity to estimate from data: a transport learned from an observed pair (ρ_k, ρ_{k+1}) both reconstructs the change already seen and, insofar as the population follows the flow (7), extrapolates the next iterate by composition. Given rolling empirical distributions, the estimated optimal-transport displacement is precisely the observed velocity $v_{\text{obs}} = (T_k - \text{id})/\Delta t$ whose alignment with the predicted Wasserstein gradient we take up in Section 5.

2.5 Mean-field games and the HJB–Fokker–Planck system

We recall the structure that links the population dynamics above to strategic behavior [??]. A representative agent with state $X_t \in \mathbb{R}^d$ chooses a control α_t with dynamics $dX_t = \alpha_t dt + \sqrt{2\nu} dB_t$ to minimize a cost depending on a running Lagrangian and on the population distribution $m_t = \text{Law}(X_t)$. At equilibrium the value function u and the density m solve the coupled system

$$\begin{cases} -\partial_t u - \nu \Delta u + H(x, \nabla u) = f(x, m), \\ \partial_t m - \nu \Delta m - \nabla \cdot (m D_p H(x, \nabla u)) = 0, \\ m(0, \cdot) = m_0, \quad u(T, \cdot) = g(x, m(T)), \end{cases} \quad (13)$$

where the first (backward) equation is the HJB equation, the second (forward) equation is the Fokker–Planck equation, and the optimal feedback is $\alpha^*(x) = -D_p H(x, \nabla u)$. As emphasized in Remark 1, the forward equation in (13) is a continuity equation, but its drift $-D_p H(x, \nabla u)$ is determined by the HJB solution and is not, in general, the Wasserstein gradient of any scalar energy.

Definition 5 (Potential mean-field game) *The game (13) is potential (variational) if the coupling derives from a functional, i.e. $f(x, m) = \frac{\delta\mathcal{G}}{\delta m}(x)$ for some $\mathcal{G}$, so that (13) arises as the optimality system of a single control-of-PDE problem. In this case the equilibrium minimizes a global energy, and the forward dynamics inherit the gradient-flow / JKO structure of Sections 2.2–2.3.*

Definition 5 is the precise hypothesis under which the “systems move toward lower energy” picture is rigorous: it is exactly when the drift can be written as $-\nabla \delta\mathcal{F}/\delta m$ that (7) and (9) apply to the population side of the game.

2.6 The mean-field market-making model

We instantiate the state x as a dealer's signed inventory and follow the strategic mean-field market-making setup of [?], in the tradition of inventory-control market making [?]. A market maker quotes bid and ask prices and accumulates inventory from the order flow generated by a continuum of market takers, whose aggregate response enters through a mean-field interaction; the equilibrium is characterized by a system of the form (13). For the variational study we focus on the induced population density m_t of inventories and posit a free energy (5) in which

- V penalizes large absolute inventory (inventory risk), e.g. $V(x) = \frac{a}{2}x^2$, capturing mean reversion of positions toward zero;
- W penalizes concentration of dealers at similar inventories (crowding), e.g. $W(x, y) = \frac{b}{2}(x - y)^2$;
- β controls entropic dispersion of inventories across the population;
- a time-dependent tilt $V_t(x)$ models exogenous liquidity shocks that transiently push the distribution away from equilibrium.

This one-dimensional specification is the toy model on which the JKO scheme of Section 2.3 will be tested against a direct Fokker–Planck solver, and whose comparative statics in (a, b, β) and response to shocks are the empirical questions of the project.

3 Main Results

We now specialize the general apparatus of Section 2 to the one-dimensional inventory model of Section 2.6 and state our results. Throughout this section we fix the quadratic specification

$$V(x) = \frac{a}{2}x^2, \quad W(x, y) = \frac{b}{2}(x - y)^2, \quad a > 0, b \geq 0, \beta > 0, \quad (14)$$

so that the free energy (5) reads, after using $\frac{1}{2} \iint W \rho \rho = \frac{b}{2} \text{Var}(\rho)$,

$$\mathcal{F}(\rho) = \frac{a}{2} \int x^2 \rho \, dx + \frac{b}{2} \text{Var}(\rho) + \beta \int \rho \log \rho \, dx. \quad (15)$$

All proofs are deferred to Appendix A.

3.1 Gradient-flow representation and its scope

The first result records exactly the sense in which the inventory dynamics “move downhill” on $\mathcal{F}$, and connects this to the potential-MFG hypothesis of Definition 5.

Proposition 6 (Gradient-flow form of the inventory dynamics) *The W_2 gradient flow (7) of the functional (15) is the nonlinear Fokker–Planck (McKean–Vlasov) equation*

$$\partial_t m_t = \nabla \cdot \left(m_t ((a + b)x - b \mu_t) \right) + \beta \Delta m_t, \quad \mu_t := \int x m_t(x) \, dx, \quad (16)$$

*equivalently the standard Fokker–Planck equation $\partial_t m_t = -\nabla \cdot (v_t m_t) + \beta \Delta m_t$ with velocity field $v_t(x) = -(a + b)x + b \mu_t$. If a market-making mean-field game of the form (13) is potential in the sense of Definition 5, with running and coupling costs chosen so that the equilibrium feedback induces the drift $-\nabla(V + W * m)$, then the forward Fokker–Planck equation of the game coincides with (16) and the population of inventories is, exactly, the Wasserstein gradient flow of $\mathcal{F}$.*

Proposition 6 is the precise content of Remark 1: the variational picture is not a metaphor once the game is potential and the drift is the gradient of a first variation, and it is *only* then. The quadratic specification (14) is the simplest instance for which the hypotheses hold and the resulting flow can be solved in closed form.

3.2 Structural guarantees of the JKO scheme

The JKO scheme (9) is attractive precisely because it transfers the qualitative structure of the continuous flow to the discrete iterates, unconditionally in the step size τ .

Theorem 7 (Well-posedness and structure preservation) *Let $\mathcal{F}$ be as in (15) and let $\rho_k \in \mathcal{P}_2(\mathbb{R})$ with $\mathcal{F}(\rho_k) < \infty$. For every $\tau > 0$:*

- (i) the JKO step (9) admits a unique minimizer $\rho_{k+1} \in \mathcal{P}_2(\mathbb{R})$, which is absolutely continuous with a strictly positive density;
- (ii) total mass and nonnegativity are preserved exactly, $\int \rho_{k+1} dx = 1$ and $\rho_{k+1} \geq 0$;
- (iii) the scheme dissipates energy at every step, with the quantitative estimate

$$\mathcal{F}(\rho_{k+1}) + \frac{1}{2\tau} W_2^2(\rho_{k+1}, \rho_k) \leq \mathcal{F}(\rho_k); \quad (17)$$

- (iv) as $\tau \rightarrow 0$, the piecewise-constant interpolant of (ρ_k) converges in $C([0, T]; \mathcal{P}_2(\mathbb{R}))$ to the unique solution of (16) with initial datum ρ_0 .

Properties (i)–(iii) are exactly the structural advantages absent from a naive explicit discretization of (16); we verify the contrast numerically in Section 4.4.

3.3 Equilibria as McKean–Vlasov–Gibbs fixed points

Proposition 8 (Equilibrium characterization) *A measure $m_\infty \in \mathcal{P}_2(\mathbb{R})$ is a stationary point of the flow (7) if and only if $\frac{\delta \mathcal{F}}{\delta \rho}(m_\infty)$ is constant on $\text{supp}(m_\infty)$; equivalently m_∞ is a McKean–Vlasov–Gibbs fixed point,*

$$m_\infty(x) = \frac{1}{Z} \exp\left(-\frac{1}{\beta}(V(x) + (W * m_\infty)(x))\right), \quad Z = \int_{\mathbb{R}} \exp\left(-\frac{1}{\beta}(V + W * m_\infty)\right). \quad (18)$$

Equation (18) is the finance reading of detailed balance: at equilibrium the inventory distribution is a Boltzmann–Gibbs density in the effective potential $V + W * m_\infty$, where the crowding term shifts the landscape self-consistently. Unlike velocity-field statements, the equilibrium is an object one can estimate robustly from data, and it is the natural target of the structure-light validation we discuss in Section 5.

3.4 The explicitly solvable Gaussian case

For the quadratic model the fixed-point equation (18) can be solved in closed form, yielding an exact equilibrium and exact relaxation rates. This is the ground truth against which the numerics of Section 4 are measured.

Theorem 9 (Gaussian equilibrium and relaxation rates) *Under (14) with $a > 0$, $b \geq 0$, $\beta > 0$:*

- (i) the flow (16) has a unique equilibrium, the centered Gaussian

$$m_\infty = \mathcal{N}\left(0, \sigma_\infty^2\right), \quad \sigma_\infty^2 = \frac{\beta}{a+b}; \quad (19)$$

- (ii) the mean μ_t and variance $\Sigma_t := \text{Var}(m_t)$ evolve in closed form,

$$\mu_t = \mu_0 e^{-at}, \quad \Sigma_t = \frac{\beta}{a+b} + \left(\Sigma_0 - \frac{\beta}{a+b}\right) e^{-2(a+b)t}, \quad (20)$$

so the mean relaxes at rate a and the variance at rate $2(a+b)$;

- (iii) $\mathcal{F}$ is a -geodesically convex on $(\mathcal{P}_2(\mathbb{R}), W_2)$, and consequently the flow is an a -contraction toward equilibrium,

$$W_2(m_t, m_\infty) \leq e^{-at} W_2(m_0, m_\infty), \quad (21)$$

while the JKO iterates contract as $W_2(\rho_k, m_\infty) \leq (1 + a\tau)^{-k} W_2(\rho_0, m_\infty)$.

Three financial readings follow immediately. First, the equilibrium dispersion $\sigma_\infty^2 = \beta/(a+b)$ decreases in the crowding strength b : the more dealers penalize being positioned like their peers, the tighter the population concentrates around flat inventory. Second, the mean inventory – the aggregate directional position of the dealer population – is the slowest mode, relaxing only at rate a set by individual inventory aversion, whereas the dispersion equilibrates faster at rate $2(a+b)$. Third, a transient liquidity shock modeled as a tilt $V_t(x) = \frac{a}{2}(x - c_t)^2$ displaces the mean toward c_t and is then erased at rate a ; we observe exactly this in Section 4.2.

4 Experiments

We test the predictions of Section 3 on the one-dimensional model and compare the JKO scheme against direct Fokker–Planck solvers. Unless stated otherwise we use $a = 1$, $b = \frac{1}{2}$, $\beta = \frac{1}{2}$, for which Theorem 9 gives $\sigma_\infty^2 = \beta/(a+b) = 1/3$, mean relaxation rate $a = 1$, and variance relaxation rate $2(a+b) = 3$.

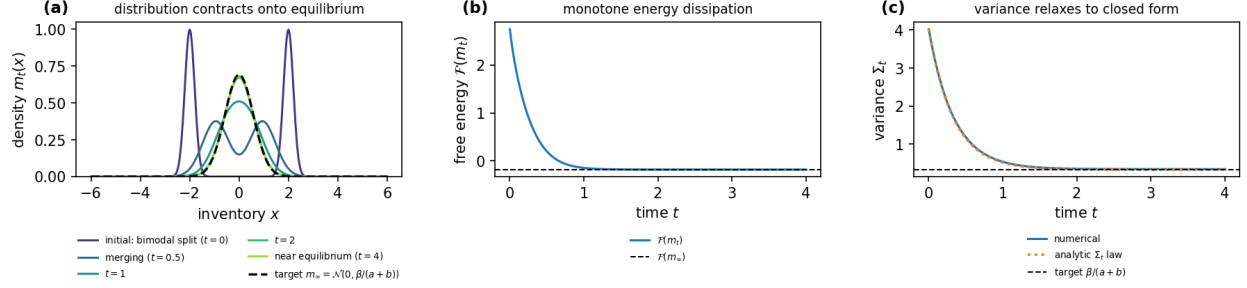


Figure 1: **Inventory relaxation.** (a) The bimodal initial distribution (dealers split between large short and long inventories) contracts onto the analytic equilibrium $\mathcal{N}(0, \beta/(a+b))$ (dashed). (b) The free energy $\mathcal{F}(m_t)$ decreases monotonically toward $\mathcal{F}(m_\infty)$ (dashed). (c) The variance follows the predicted exponential law $\Sigma_t = \frac{\beta}{a+b} + (\Sigma_0 - \frac{\beta}{a+b})e^{-2(a+b)t}$ (dotted) toward $\beta/(a+b)$ (dashed).

Numerical methods. The *Eulerian* solvers discretize (16) on a uniform grid of the inventory axis $[-6, 6]$ ($N = 481$ nodes, $\Delta x \approx 0.025$) in conservative flux form with upwind advection, central diffusion, and reflecting (no-flux) boundaries; we use both an explicit (forward-Euler) and an implicit (backward-Euler) time stepping. The *JKO* solver works in the 1-dimensional Lagrangian coordinate given by the quantile function $Q = F_\rho^{-1}$, in which the squared 2-Wasserstein distance is the flat $L^2(0, 1)$ distance $W_2^2(\rho, \rho_k) = \int_0^1 (Q - Q_k)^2 du$ and the energy reads $\mathcal{F}[Q] = \frac{a}{2} \int Q^2 + \frac{b}{2} \text{Var}[Q] - \beta \int_0^1 \log Q' du$ [?]. Each JKO step (9) is then a smooth convex program in Q over the cone of nondecreasing functions ($M = 240$ quantile nodes), solved by L-BFGS; this exactly enforces mass conservation and positivity. For higher-dimensional state one would instead use entropic/Sinkhorn proximal steps [??] or neural JKO parametrizations [?]; the 1D quantile scheme suffices for the present toy model.

4.1 Inventory relaxation

We initialize a bimodal inventory distribution $m_0 = \frac{1}{2}\mathcal{N}(-2, 0.2^2) + \frac{1}{2}\mathcal{N}(2, 0.2^2)$, representing a dealer population split between large long and large short positions, and evolve (16) to $T = 4$ ($\Delta t = 2 \times 10^{-3}$, implicit). Figure 1 shows the distribution contracting onto the predicted equilibrium $\mathcal{N}(0, 1/3)$. The free energy decreases monotonically to $\mathcal{F}(m_\infty)$, with no energy-increasing steps, and the variance tracks the analytic curve $\Sigma_t = \frac{1}{3} + (\Sigma_0 - \frac{1}{3})e^{-3t}$ of Theorem 9(ii); the terminal variance is 0.345 against the theoretical 0.333, the small excess being numerical diffusion of the Eulerian scheme.

4.2 Liquidity shock

Starting from equilibrium, we apply a transient tilt of the potential, $V_t(x) = \frac{a}{2}(x - c_t)^2$ with $c_t = 2$ on the window $t \in [2, 4]$ and $c_t = 0$ otherwise, modeling a temporary directional pressure on dealer inventories. Figure 2 shows the mean inventory μ_t moving toward the tilt target during the shock and decaying back afterward at rate a . Consistent with Theorem 9, the mean reaches 1.72 at the end of the shock window, matching the quasi-static prediction $c(1 - e^{-a\Delta T}) = 2(1 - e^{-2}) \approx 1.73$; the Wasserstein distance to equilibrium departs and returns on the same $1/a$ timescale. The energy landscape, not the price, is what the shock perturbs, and the distribution relaxes back along the gradient flow.

4.3 Crowding sweep

We vary the crowding strength b over $[0, 3]$ and compute the numerical equilibrium variance, comparing it to $\beta/(a+b)$. Figure 3 confirms the predicted compression of the inventory distribution as crowding increases, with a maximum absolute error of 1.4×10^{-2} across the sweep. At $b = 0$ dealers act independently and the equilibrium is the widest; increasing b pulls the population toward a common (zero) inventory and narrows the stationary distribution, the comparative static promised after Theorem 9.

4.4 Solver comparison

Finally we compare the three solvers at a deliberately large step. On a coarse grid ($N = 161$, $\Delta x = 0.075$) the explicit Fokker–Planck scheme is stable only for $\Delta t < \Delta x^2/2\beta \approx 5.6 \times 10^{-3}$; we run all methods at $\Delta t = 1.8 \times$ this limit (1.0×10^{-2}), and the JKO scheme at the comparable proximal step $\tau = 5 \times 10^{-2}$. Figure 4 and Table 1 report the

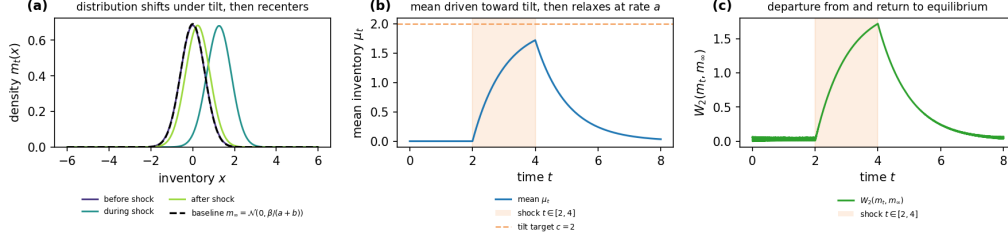


Figure 2: **Liquidity shock** (shaded window $t \in [2, 4]$, tilt $c = 2$). (a) The equilibrium distribution shifts rightward under the tilted potential, then recenters toward the baseline $\mathcal{N}(0, \beta/(a+b))$ (dashed) after the shock ends. (b) The mean inventory μ_t is driven toward the tilt target and relaxes back toward zero at rate a . (c) Departure from and return to equilibrium, measured by $W_2(m_t, m_\infty)$.

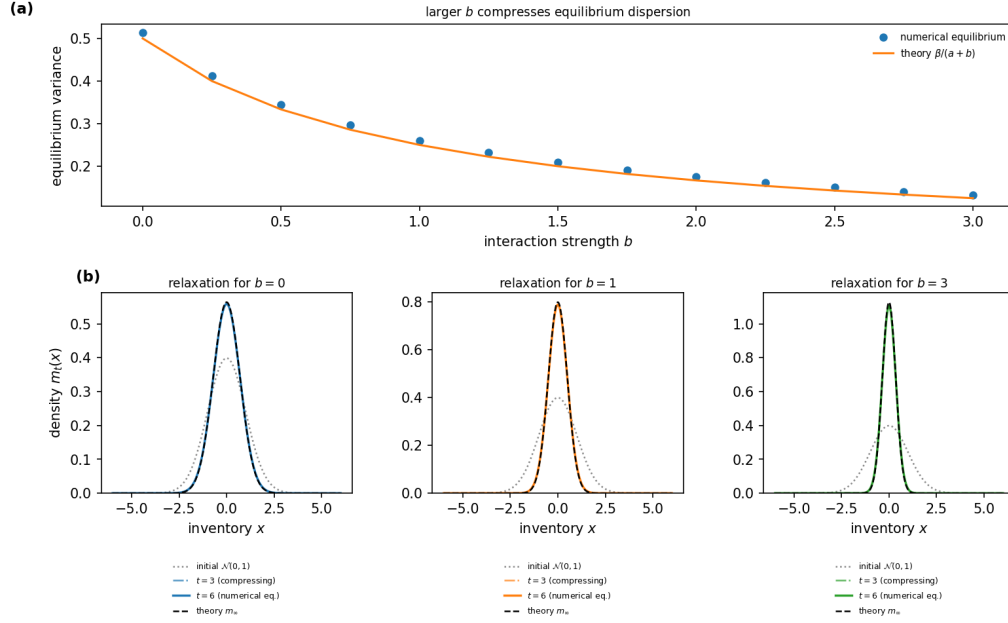


Figure 3: **Crowding sweep**. (a) Numerical equilibrium variance (markers) against the theoretical $\beta/(a+b)$ (line): larger interaction strength b compresses dispersion. (b) Relaxation from a common wide initial $\mathcal{N}(0, 1)$ to equilibrium for $b \in \{0, 1, 3\}$; stronger crowding yields a tighter terminal distribution (solid) matching theory m_∞ (dashed).

outcome. The explicit scheme develops negative mass and an exploding energy and diverges by $t \approx 0.61$; the implicit scheme and the JKO scheme remain stable, conserve mass, preserve positivity, and dissipate energy monotonically, exactly as Theorem 7 guarantees for JKO. At equal (coarse) resolution the JKO equilibrium variance (0.345) is closer to the analytic 0.333 than the upwind implicit solver (0.381), whose numerical diffusion inflates the dispersion – the interpretability and accuracy dividend of respecting the Wasserstein structure.

5 Conclusion

We have made precise, in a one-dimensional mean-field market-making model, the sense in which a distribution of dealer inventories “moves toward lower energy.” Under a quadratic inventory potential, a quadratic crowding interaction, and an entropic dispersion term, the population density is the 2-Wasserstein gradient flow of an explicit free-energy functional (Proposition 6); the JKO scheme inherits mass conservation, positivity, and monotone energy dissipation unconditionally in the step size (Theorem 7); equilibria are McKean–Vlasov–Gibbs densities in the self-consistent effective potential (Proposition 8); and in the Gaussian case the equilibrium $\mathcal{N}(0, \beta/(a+b))$ and the relaxation rates a and $2(a+b)$ are available in closed form (Theorem 9). The experiments confirm each prediction and show that the JKO discretization is both more stable and, at equal resolution, more accurate than a direct upwind Fokker–Planck solver.

Table 1: Solver comparison at the large step $\Delta t = 1.8 \Delta t_{\text{CFL}}$ (JKO at $\tau = 5 \times 10^{-2}$). “Energy $\uparrow$ steps” counts iterations that increase $\mathcal{F}$; equilibrium variance is compared to the analytic $\beta/(a+b) = 0.333$.

Method	Stable at large step	Energy $\uparrow$ steps	Mass error	Positivity	Eq. variance
Explicit FD	no (diverges $t \approx 0.61$)	32	$\rightarrow \infty$	violated	—
Implicit FD	yes	0	2.2×10^{-16}	preserved	0.381
JKO (quantile)	yes	0	0 (exact)	exact	0.345

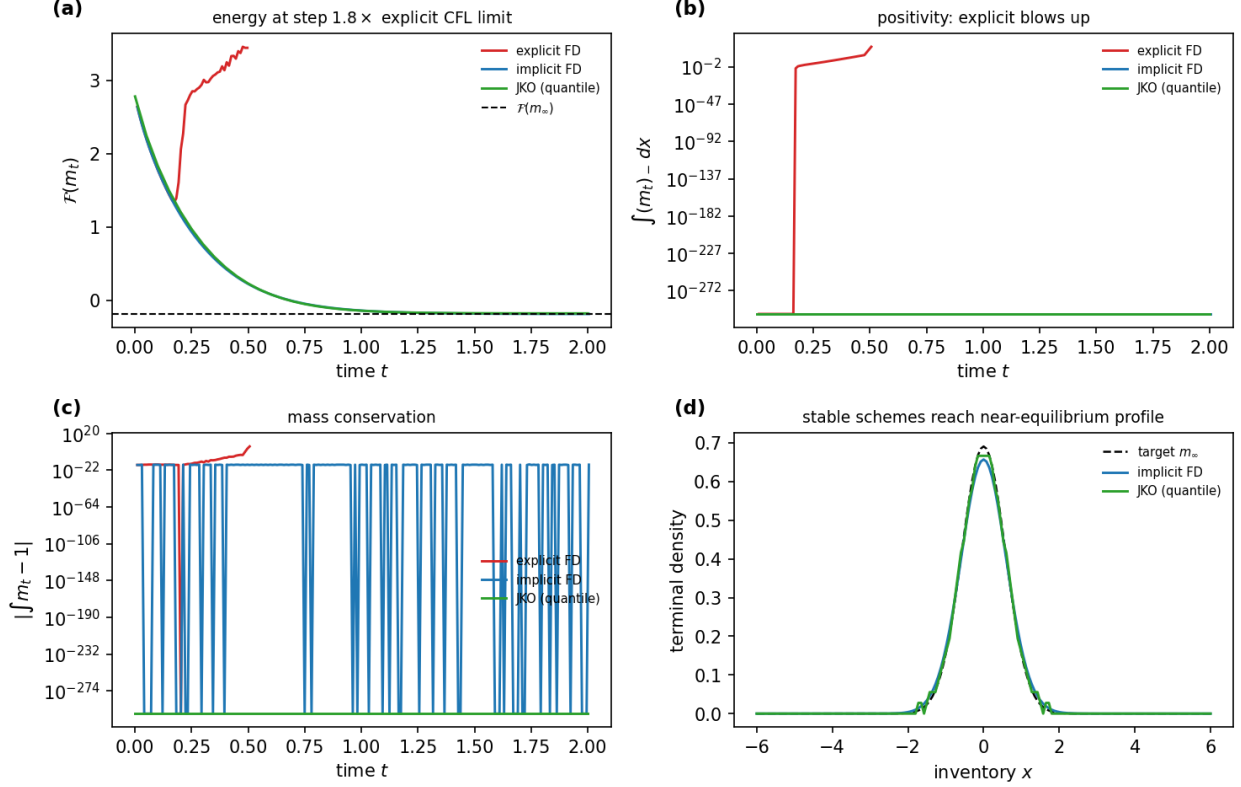


Figure 4: **Solver comparison** at a step $1.8 \times$ the explicit stability limit. (a) Energy: the explicit scheme (red) diverges; implicit and JKO decrease monotonically toward $\mathcal{F}(m_\infty)$ (dashed). (b) Negative mass $\int (m_t)_- dx$: explicit blows up; implicit and JKO remain at machine zero (log scale). (c) Mass-conservation error $|\int m_t - 1|$: same pattern. (d) Terminal densities for the stable schemes versus the target equilibrium (dashed).

Limitations. The model is deliberately stylized. Dealer inventories are not directly observable at the population level; the quadratic specification is the one that makes the gradient-flow structure exact, and richer costs need not be potential; and, as stressed in Remark 1, a general market-making MFG is *not* a Wasserstein gradient flow. The closed-form Gaussian results rely on the quadratic interaction, and the 1D quantile solver does not extend directly to high-dimensional agent states.

Toward empirical validation. The natural next step is to ask whether *observed* market-state distributions move in the direction the energy predicts. Because a gradient flow and any positive time-reparametrization trace the same path, the step-size-free content of the hypothesis is directional: given rolling empirical distributions $\hat{m}_k \rightarrow \hat{m}_{k+1}$, one estimates the optimal-transport velocity $v_{\text{obs}} = (T_k - \text{id})/\Delta t$ and measures its alignment, in the $L^2(\hat{m}_k)$ (Otto) inner product, with the predicted Wasserstein gradient $-\nabla \delta \mathcal{F}_\theta / \delta m$. We expect real dynamics to be only partially gradient-like – markets are non-equilibrium steady states with persistent circulation – so the honest target is to *quantify* the gradient fraction and the non-gradient (rotational) residual, rather than to assert a pure gradient flow. Complementary structure-light tests include monitoring monotone decrease of a fitted $\mathcal{F}$ as a Lyapunov function, checking the Gibbs form (18) of the empirical stationary distribution, and reconstructing the drift directly from conditional increments in the spirit of Kramers–Moyal estimation [?]; Wasserstein distances between rolling windows also give a robust,

assumption-light regime monitor [?]. Learning a parametric energy $\mathcal{F}_\theta$ by maximizing directional alignment, then testing out-of-sample, is left to future work.

References

A Proofs

We collect the proofs of the results of Section 3. Throughout, $\mathcal{F}$ is the quadratic free energy (15) on $\mathcal{P}_2(\mathbb{R})$, and we write $\mu(\rho) = \int x \rho dx$ and $\Sigma(\rho) = \int x^2 \rho dx - \mu(\rho)^2$ for the mean and variance.

A.1 Proof of Proposition 6

We first compute the first variation of (15). For the interaction term, write $\frac{1}{2} \iint W(x, y) \rho(x) \rho(y) = \frac{b}{2} \iint \frac{1}{2} (x - y)^2 \rho(x) \rho(y) dx dy$; perturbing $\rho \mapsto \rho + \varepsilon \chi$ with $\int \chi = 0$ and collecting the $O(\varepsilon)$ term, the symmetric kernel contributes $\int (W * \rho)(x) \chi(x) dx$, so that the interaction first variation is $W * \rho$. Together with the elementary variations of the potential and entropy terms,

$$\frac{\delta \mathcal{F}}{\delta \rho}(x) = V(x) + (W * \rho)(x) + \beta(\log \rho(x) + 1), \quad (22)$$

in agreement with (6). For the quadratic data (14),

$$(W * \rho)(x) = \int \frac{b}{2} (x - y)^2 \rho(y) dy = \frac{b}{2} \left(x^2 - 2x \mu(\rho) + \int y^2 \rho dy \right), \quad (23)$$

whence $\nabla(W * \rho)(x) = b(x - \mu(\rho))$ and $\nabla V(x) = ax$. Therefore

$$\nabla \frac{\delta \mathcal{F}}{\delta \rho}(x) = ax + b(x - \mu(\rho)) + \beta \frac{\nabla \rho(x)}{\rho(x)} = ((a + b)x - b\mu(\rho)) + \beta \frac{\nabla \rho}{\rho}. \quad (24)$$

Substituting into the gradient-flow equation (7),

$$\partial_t m_t = \nabla \cdot \left(m_t \nabla \frac{\delta \mathcal{F}}{\delta \rho}(m_t) \right) = \nabla \cdot \left(m_t ((a + b)x - b\mu_t) \right) + \beta \nabla \cdot (\nabla m_t), \quad (25)$$

which is (16), since $\nabla \cdot \nabla = \Delta$. Writing this in the form $\partial_t m_t = -\nabla \cdot (v_t m_t) + \beta \Delta m_t$ identifies the velocity $v_t(x) = -(a + b)x + b\mu_t$.

For the mean-field-game statement, recall from Definition 5 that in a potential game the coupling is a first variation, $f(x, m) = \delta \mathcal{G} / \delta m$, and the forward equation of (13) is the continuity equation with drift $-D_p H(x, \nabla u)$ determined by the optimality system. If the running cost and the Hamiltonian are chosen so that the equilibrium feedback reproduces the drift $-\nabla(V + W * m)$ – as is the case for a quadratic (linear-quadratic) inventory cost with the mean-field penalty $\frac{1}{2} \iint W mm$ – then the forward equation is precisely $\partial_t m = \nabla \cdot (m \nabla(V + W * m)) + \beta \Delta m$, i.e. (16). $\square$

A.2 Proof of Theorem 7

(i) *Existence and uniqueness.* The map $\rho \mapsto \frac{1}{2\tau} W_2^2(\rho, \rho_k)$ is lower semicontinuous for the narrow topology and has W_2 -bounded sublevel sets; $\mathcal{F}$ is lower semicontinuous and bounded below on $\mathcal{P}_2(\mathbb{R})$ (the entropy is controlled by the second moment via the Gaussian, and $V, W \geq 0$). Hence the JKO objective is l.s.c. with compact sublevel sets and a minimizer exists by the direct method [??]. For uniqueness, by Theorem 9(iii) the functional $\mathcal{F}$ is a -geodesically convex with $a > 0$, and $\rho \mapsto \frac{1}{2\tau} W_2^2(\rho, \rho_k)$ is $\frac{1}{\tau}$ -geodesically convex along generalized geodesics; the sum is strictly (indeed $(a + \frac{1}{\tau})$ -)convex, so the minimizer is unique [?, Ch. 4].

Absolute continuity and positivity. If ρ_{k+1} had a singular part, replacing it by its absolutely continuous regularization strictly decreases the entropy $\int \rho \log \rho$ while changing $W_2^2(\cdot, \rho_k)$ and the other terms by an arbitrarily small amount, contradicting minimality; hence $\rho_{k+1} \ll \text{Leb}$. The optimality condition for (9) reads, with φ_k the Kantorovich potential from ρ_{k+1} to ρ_k ,

$$\frac{\delta \mathcal{F}}{\delta \rho}(\rho_{k+1})(x) + \frac{\varphi_k(x)}{\tau} = \text{const} \quad \text{on } \text{supp}(\rho_{k+1}). \quad (26)$$

Inserting (22) and solving for the density,

$$\rho_{k+1}(x) = \exp \left(\frac{1}{\beta} (C - V(x) - (W * \rho_{k+1})(x) - \tau^{-1} \varphi_k(x)) - 1 \right) > 0, \quad (27)$$

so the density is strictly positive wherever defined.

(ii) *Mass and positivity.* The minimization is over $\mathcal{P}_2(\mathbb{R})$, so ρ_{k+1} is by definition a probability measure: $\int \rho_{k+1} = 1$ and $\rho_{k+1} \geq 0$. No discretization of the flux is involved, so these hold exactly.

(iii) *Energy dissipation.* Since ρ_{k+1} minimizes the objective and ρ_k is admissible with $W_2(\rho_k, \rho_k) = 0$,

$$\mathcal{F}(\rho_{k+1}) + \frac{1}{2\tau} W_2^2(\rho_{k+1}, \rho_k) \leq \mathcal{F}(\rho_k) + \frac{1}{2\tau} W_2^2(\rho_k, \rho_k) = \mathcal{F}(\rho_k), \quad (28)$$

which is (17); in particular $\mathcal{F}(\rho_{k+1}) \leq \mathcal{F}(\rho_k)$.

(iv) *Convergence as $\tau \rightarrow 0$.* Estimate (17) is the discrete energy-dissipation inequality of the minimizing-movements (De Giorgi) scheme. Summing it yields the total-square-distance bound $\sum_k W_2^2(\rho_{k+1}, \rho_k) \leq 2\tau(\mathcal{F}(\rho_0) - \inf \mathcal{F})$, which gives equicontinuity (a uniform $\frac{1}{2}$ -Hölder bound) of the piecewise-constant interpolants; by the refined Ascoli theorem in $(\mathcal{P}_2, W_2)$ a subsequence converges uniformly to an absolutely continuous curve, and the a -convexity of $\mathcal{F}$ identifies the limit as the unique gradient flow / curve of maximal slope of (16) [?, Thm. 11.1.6 and Thm. 4.0.4]; uniqueness of the limit promotes convergence of the whole family. $\square$

A.3 Proof of Proposition 8

A measure m_∞ is stationary for (7) iff the right-hand side vanishes, i.e. $\nabla \cdot (m_\infty \nabla \delta \mathcal{F} / \delta \rho(m_\infty)) = 0$. Testing against $\delta \mathcal{F} / \delta \rho(m_\infty)$ and integrating by parts,

$$0 = - \int m_\infty \left| \nabla \frac{\delta \mathcal{F}}{\delta \rho}(m_\infty) \right|^2 dx, \quad (29)$$

forces $\nabla \delta \mathcal{F} / \delta \rho(m_\infty) = 0$ m_∞ -a.e., i.e. $\delta \mathcal{F} / \delta \rho(m_\infty)$ is constant on $\text{supp}(m_\infty)$. Conversely a constant first variation makes the drift vanish. Setting (22) equal to a constant C and solving,

$$\beta(\log m_\infty + 1) = C - V - W * m_\infty \implies m_\infty(x) = \frac{1}{Z} \exp\left(-\frac{1}{\beta}(V(x) + (W * m_\infty)(x))\right), \quad (30)$$

with Z the normalizing constant, which is (18). $\square$

A.4 Proof of Theorem 9

(i) *Equilibrium.* For the quadratic data, $(W * m_\infty)(x) = \frac{b}{2}(x^2 - 2x\mu_\infty + \int y^2 m_\infty)$ with $\mu_\infty = \mu(m_\infty)$. The effective potential in (18) is therefore quadratic in x ,

$$V(x) + (W * m_\infty)(x) = \frac{a+b}{2}x^2 - b\mu_\infty x + \text{const}, \quad (31)$$

so m_∞ is Gaussian. Completing the square, its mean is $b\mu_\infty/(a+b)$; self-consistency $\mu_\infty = b\mu_\infty/(a+b)$ forces $\mu_\infty(a/(a+b)) = 0$, hence $\mu_\infty = 0$ since $a > 0$. The coefficient of x^2 is $(a+b)/(2\beta)$, giving variance $\sigma_\infty^2 = \beta/(a+b)$ and proving (19). Uniqueness follows from Proposition 8 together with strict convexity.

(ii) *Moment dynamics.* Multiply (16) in the form $\partial_t m = -\nabla \cdot (v m) + \beta \Delta m$, $v(x) = -(a+b)x + b\mu_t$, by x and integrate; using $\int x \partial_x(\cdot) = -\int(\cdot)$ and the decay of m at infinity,

$$\dot{\mu}_t = \int x \partial_t m_t dx = \int v m_t dx = -(a+b)\mu_t + b\mu_t = -a\mu_t, \quad (32)$$

so $\mu_t = \mu_0 e^{-at}$. Multiplying instead by x^2 and writing $S_t = \int x^2 m_t$,

$$\dot{S}_t = \int 2x v m_t dx + 2\beta = -2(a+b)S_t + 2b\mu_t^2 + 2\beta. \quad (33)$$

With $\Sigma_t = S_t - \mu_t^2$ and $(\dot{\mu}_t^2) = -2a\mu_t^2$,

$$\dot{\Sigma}_t = \dot{S}_t - 2\mu_t \dot{\mu}_t = -2(a+b)S_t + 2b\mu_t^2 + 2\beta + 2a\mu_t^2 = -2(a+b)\Sigma_t + 2\beta, \quad (34)$$

where the μ_t^2 terms cancel because $-2(a+b) + 2b + 2a = 0$. Integrating gives the variance law in (20), with equilibrium $\beta/(a+b)$ and rate $2(a+b)$.

(iii) *Displacement convexity and contraction.* By McCann's criterion [??], on $(\mathcal{P}_2(\mathbb{R}), W_2)$ the internal energy $\beta \int \rho \log \rho$ is geodesically convex (convexity constant 0); the potential energy $\int V \rho$ is λ_V -geodesically convex whenever V is λ_V -convex, here $\lambda_V = a$ since $V'' = a$; and the interaction energy $\frac{1}{2} \iint W(x-y) \rho \rho$ with $W(z) = \frac{b}{2} z^2$ convex is geodesically convex (nonnegative contribution). Summing, $\mathcal{F}$ is a -geodesically convex. The general theory of gradient flows of λ -convex functionals then yields the exponential contraction [?, Thm. 11.2.1], $W_2(m_t, m_\infty) \leq e^{-at} W_2(m_0, m_\infty)$, and the corresponding contraction of the resolvent (JKO) operator, $W_2(\rho_{k+1}, m_\infty) \leq (1 + a\tau)^{-1} W_2(\rho_k, m_\infty)$, whence the stated $(1 + a\tau)^{-k}$ bound; see also ? and the Bakry-Émery viewpoint [?]. The rate is sharp: by (20) the mean is the slowest mode, relaxing at exactly a while the variance relaxes faster at $2(a+b)$. $\square$

B On the one-dimensional JKO discretization

In one dimension the optimal transport between ρ and ρ_k is the monotone rearrangement, and $W_2^2(\rho, \rho_k) = \int_0^1 (Q_\rho(u) - Q_{\rho_k}(u))^2 du$ where $Q_\rho = F_\rho^{-1}$ is the quantile function [?]. Pushing the free energy through this change of variable, using $\rho(Q(u)) = 1/Q'(u)$, gives

$$\mathcal{F}[Q] = \frac{a}{2} \int_0^1 Q(u)^2 du + \frac{b}{2} \left(\int_0^1 Q^2 - \left(\int_0^1 Q \right)^2 \right) - \beta \int_0^1 \log Q'(u) du, \quad (35)$$

so that each JKO step is the strictly convex program $\min_{Q \nearrow} \mathcal{F}[Q] + \frac{1}{2\tau} \int_0^1 (Q - Q_k)^2 du$ over nondecreasing Q . We discretize u on $M = 240$ midpoint nodes, parametrize monotonicity through nonnegative increments, and solve with L-BFGS. Mass is conserved and positivity ($Q' > 0$) is enforced exactly by construction, which is why the JKO column of Table 1 attains machine-zero mass error and exact positivity. In higher dimensions the monotone rearrangement is unavailable and one resorts to entropic regularization of the proximal step [??], with the general computational toolbox of ?, or to neural parametrizations of the JKO step [?].